East West University

Department of Computer Science and Engineering (CSE)
Semester: (Summer, Year: 2025), B.Sc. in C SE

Smart Hospital Network Infrastructure with Centralized Services using Cisco Packet Tracer

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Course Title: ***CSE405***
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Chapter 1: Introduction

1.1 Overview

This project designs a secure, scalable smart hospital network using Cisco Packet Tracer. It connects departments and administration under centralized services, including FTP with role-based access and email communication.

1.2 Motivation

The project aims to simulate a real-world hospital network that enhances efficiency, security, and interdepartmental communication while supporting centralized digital services.

1.3 Objectives

To design and implement a secure, scalable hospital-wide network infrastructure for a smart healthcare system using Cisco Packet Tracer. The network will seamlessly support patient record management, real-time medical device monitoring, and efficient departmental communication, while ensuring prioritized delivery of critical services.

Chapter 2: Design/Development/Implementation of the Project

2.1 Project Details

A visual topology diagram illustrating the entire infrastructure, displaying the routers, switches, departments, and servers is displayed below:

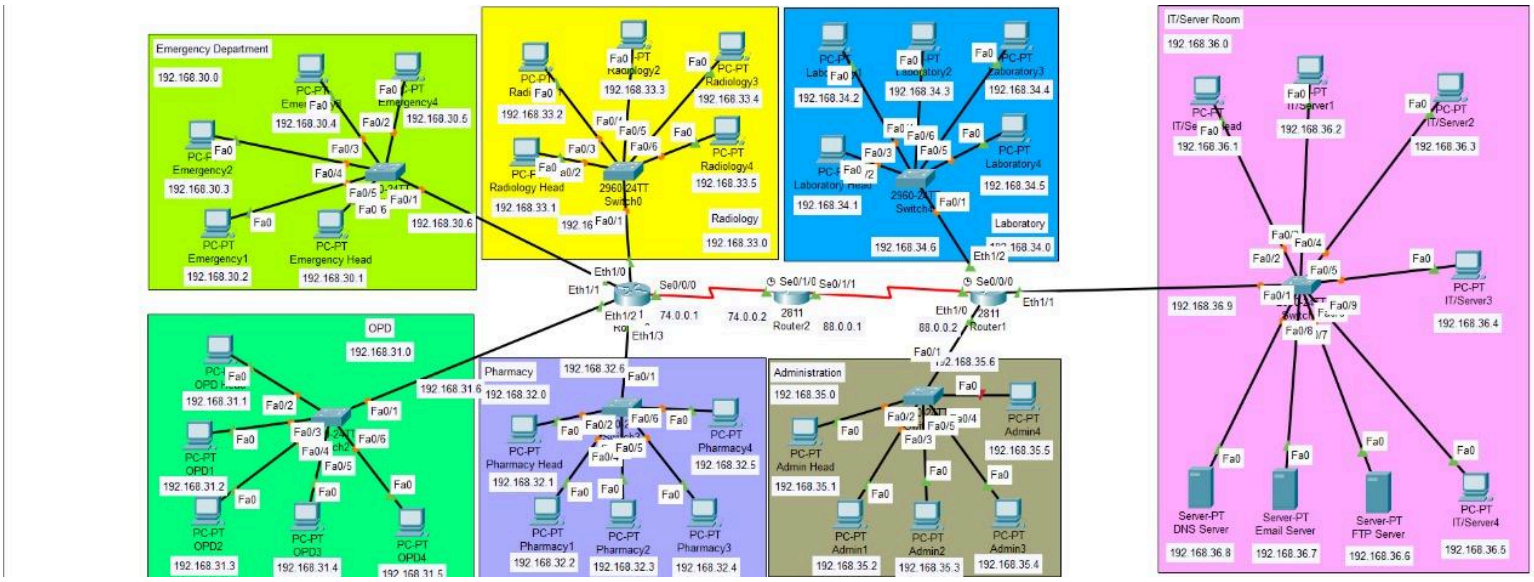

Figure: Departmental Network Topology of East West University

The network is structured as a hierarchical campus network, where individual departments function as independent LANs connected via core routers. Each LAN contains user PCs and a local switch, while interdepartmental connectivity is handled through three interconnected routers using serial links. The admin section serves as a centralized service hub.

Topology Layers and Components:

- **Core Routers (Router Layer):**
  - Five 2811 Routers interconnect the departmental LANs using serial interfaces.
  - The routers form a mesh-like connection, facilitating robust communication between department subnets.
  - Serial links are used for establishing connectivity between routers

2.2 Implementation

2.2.1 IP Configuration

A structured IPv4 addressing plan was devised to ensure clarity and ease of management:

Department	Subnet Address	Gateway	Assigned IPs
Emergency Department	192.168.30.0/24	192.168.30.6	192.168.30.1 - 192.168.30.4
Outpatient Department(OPD)	192.168.31.0/24	192.168.31.6	192.168.31.1 - 192.168.31.4
Pharmacy Department	192.168.32.0/24	192.168.32.6	192.168.32.1 - 192.168.32.5
Radiology Department	192.168.33.0/24	192.168.33.6	192.168.33.1 - 192.168.33.5
Laboratory Department	192.168.34.0/24	192.168.34.6	192.168.34.1 - 192.168.34.5
Administration section	192.168.35.0/24	192.168.35.6	192.168.35.1 - 192.168.35.5
It/Server Room	192.168.36.0/24	192.168.36.9	192.168.36.1 - 192.168.36.9

Assign IP addresses to all servers and PCs as per the IP Addressing Table.

❖ For Routers:

- Click on Router → CLI
- Type: **Router>**enable

  Router#configure terminal
  Router(config)#interface [*Interface_Type]*
  Router(config-if)#ip address [*IP_Address] [Subnet_Mask]*
  Router(config-if)#no shutdown
  Router(config-if)#exit

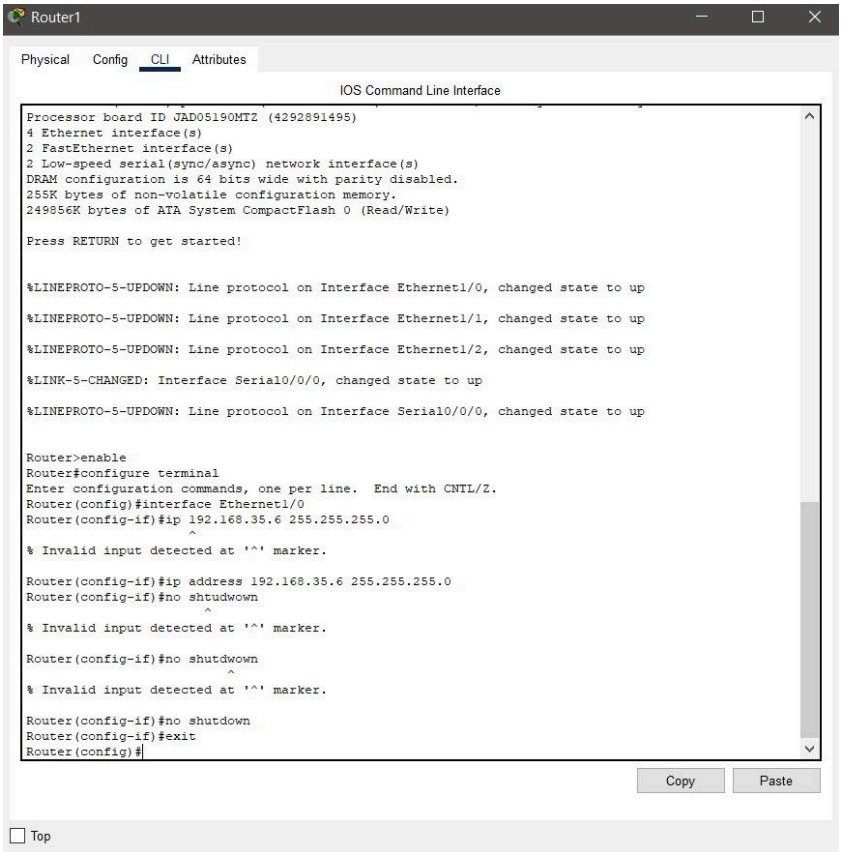

Figure: IP assignment of Router

❖ For PCs & Routers:

- Click on PC or Server→ Desktop → IP Configuration
- Assign appropriate IPv4 Address, Subnet Mask, Default Gateway according to the IP Addressing Table.

❖ **Screenshot:**

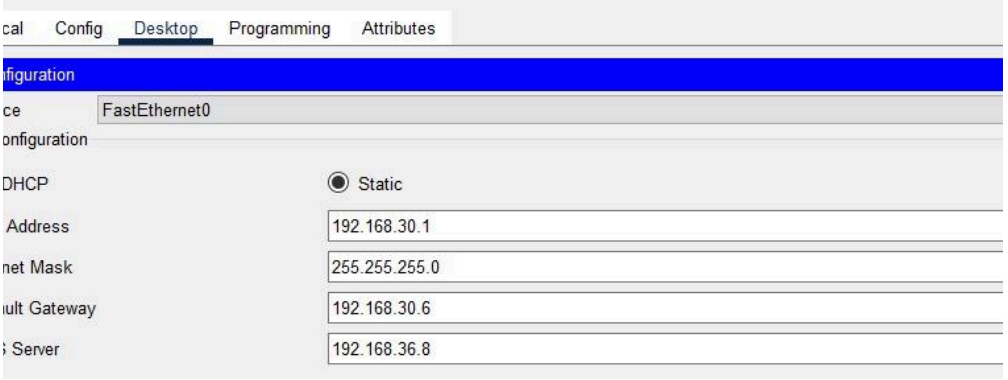

Figure: IP assignment of PC

2.2.2 FTP Server Setup:

❖ **Screenshot:**

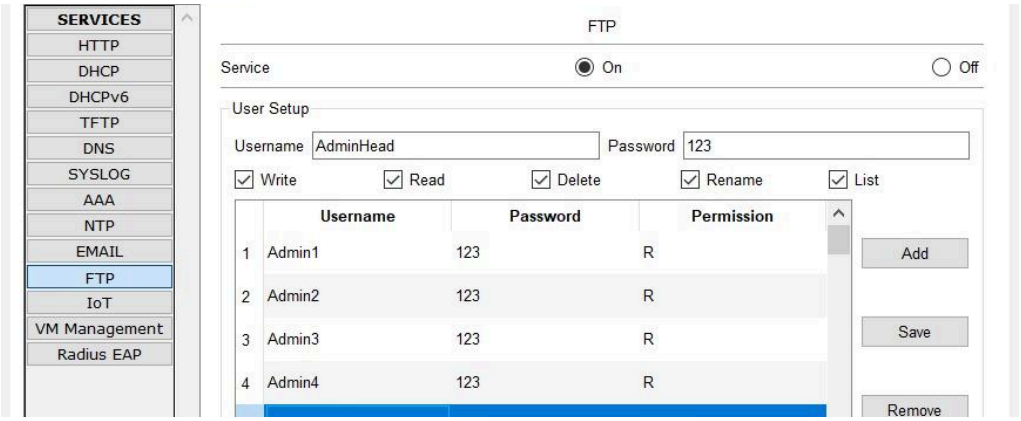

Figure: Configuration of FTP Server

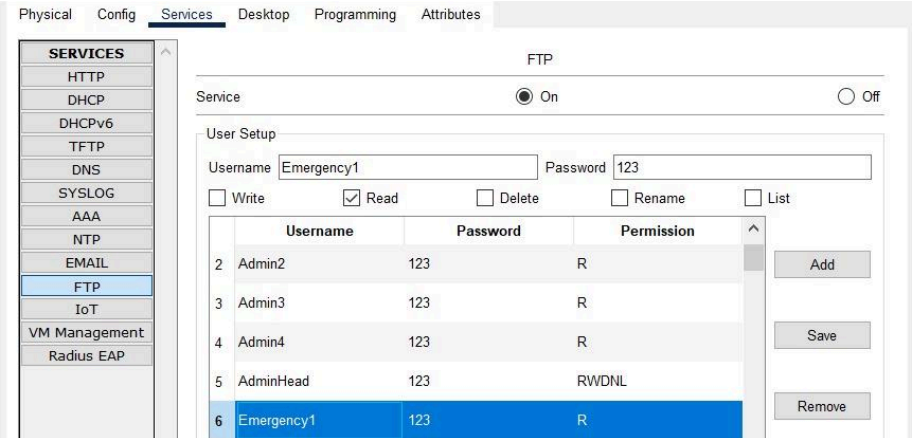

Figure: Configuration of FTP Server for PC Admin

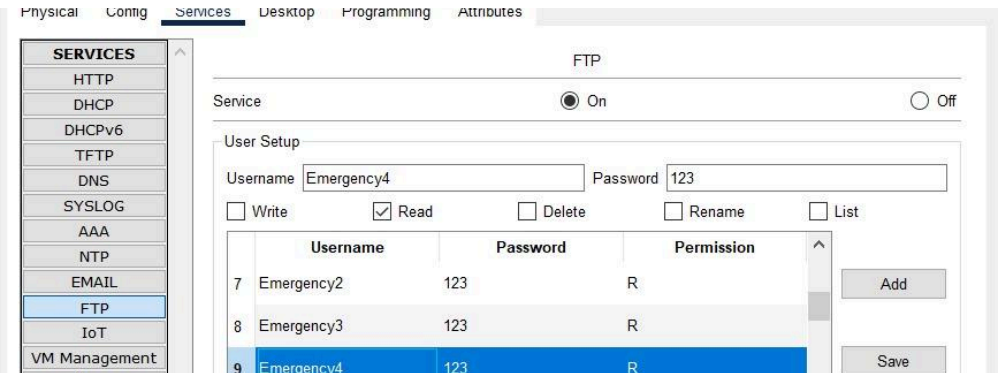

Figure: Configuration of FTP Server

2.2.3 Email & DNS Server Configuration:

❖ **PC Input/Output setup:**

- Click any PC and click the Desktop tab > Email.
- Assign appropriate name, Email address, incoming Mail Server, outgoing Mail Server, User Name, Password and click Save.

❖ **Screenshot:**

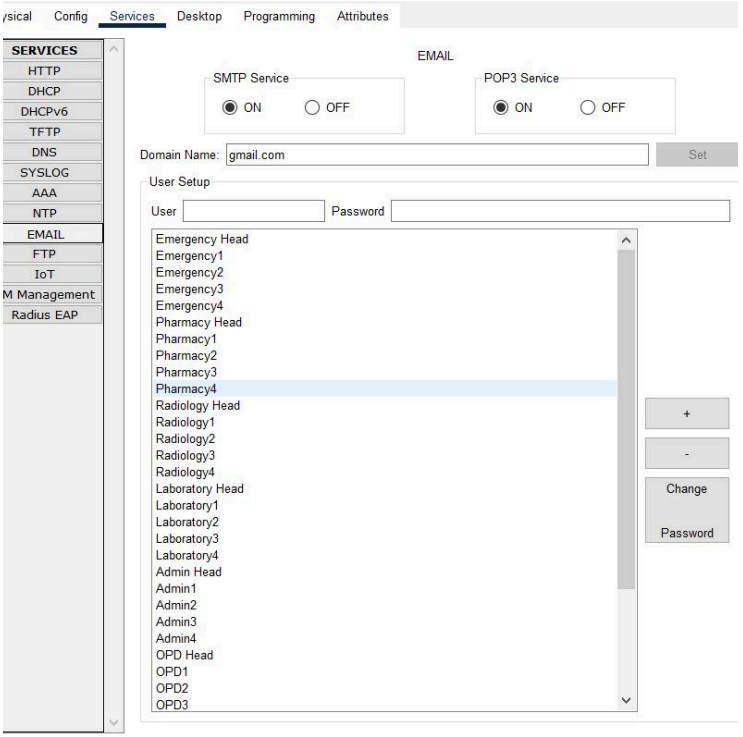

Figure: Configuration of Email Server

- **DNS Server:**

Figure: Configuration of DNS Server

- Input/Output setup:

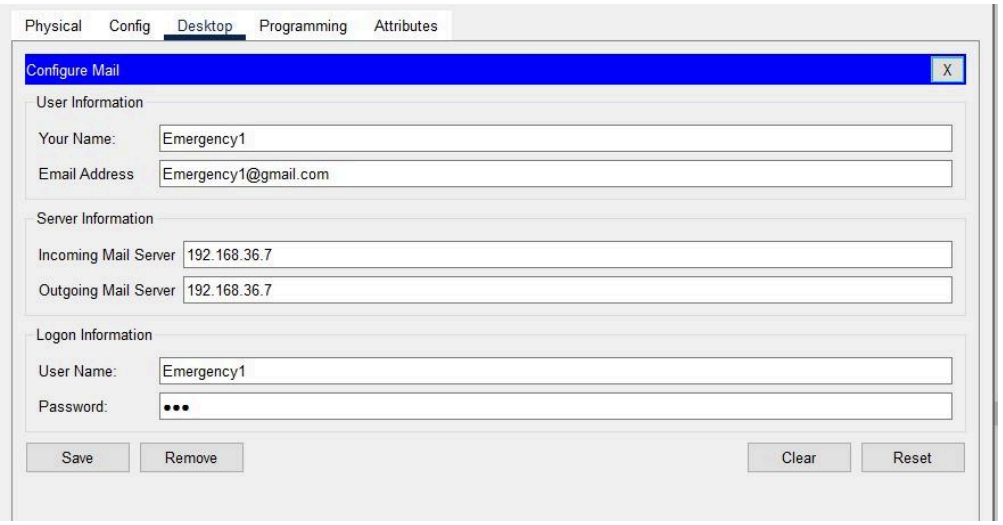

Figure: Configuration of Email service of the Mail server

2.2.4 Routing Configuration:

- Click on Router → CLI
- Type: **Router>**enable

  Router#configure terminal
  Router(config)# router rip
  Router(config-router)#network *[Network Address]*
  Router(config-router)#exit

❖ **Screenshot:**

```
Router con0 is now available


Press RETURN to get started.


Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 192.168.30.0
Router(config-router)#network 192.168.31.0
Router(config-router)#network 192.168.32.0
Router(config-router)#network 192.168.33.0
Router(config-router)#network 192.168.34.0
Router(config-router)#network 192.168.35.0
Router(config-router)#network 192.168.36.0
Router(config-router)#
```

Figure: Dynamic Routing Configuration

Chapter 3: Result and Discussion

3.1 Network Connectivity:

A comprehensive set of connectivity tests were conducted to verify the network's functionality. Using the ping command, each PC was able to communicate successfully with others both within and outside its subnet. This confirmed proper IP assignments and functional static routing configurations across all routers.

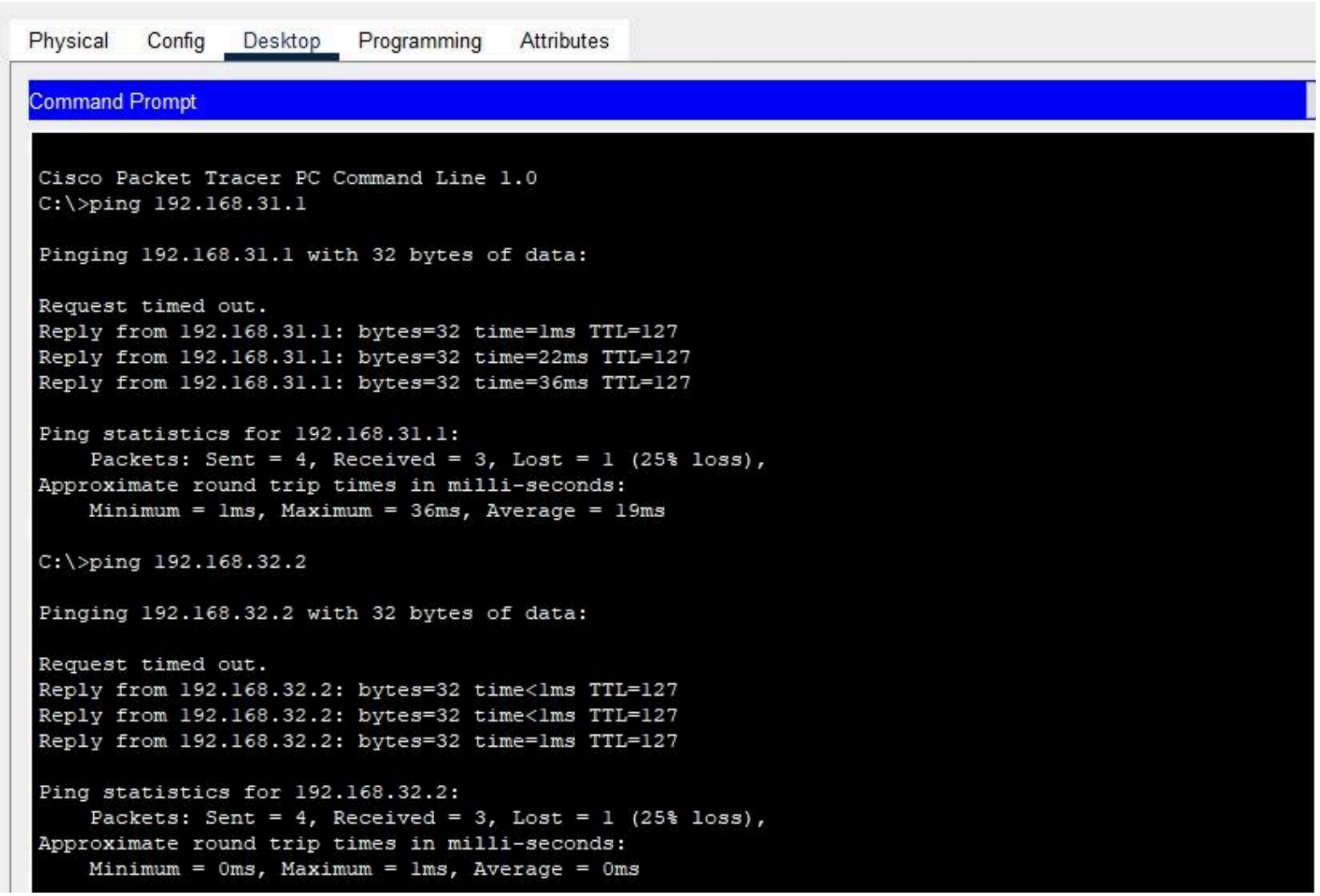

Figure : connectivity check (pinging between networks)

```
Pinging 192.168.34.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.34.2: bytes=32 time=53ms TTL=125
Reply from 192.168.34.2: bytes=32 time=57ms TTL=125
Reply from 192.168.34.2: bytes=32 time=4ms TTL=125

Ping statistics for 192.168.34.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 57ms, Average = 38ms

C:\>
```

3.2 FTP Access Testing:

The FTP server was accessed using multiple user profiles. Chairman and Admin users were tested for full access privileges, including uploading, renaming, and deleting files, which succeeded. Other department PCs were tested for read-only access, verifying that access restrictions were appropriately enforced via FTP user policies.

- **Screenshot:**

```
C:\>delete abc.txt
C:\>ftp 192.168.36.6
Trying to connect...192.168.36.6
Connected to 192.168.36.6
220- Welcome to PT Ftp server
Username:EmergencyHead
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>put EmergencyHead
%Error opening c:EmergencyHead (No such file or directory)
ftp>put EmergencyHead.txt

Writing file EmergencyHead.txt to 192.168.36.6:
File transfer in progress...

[Transfer complete - 14 bytes]

14 bytes copied in 0.019 secs (736 bytes/sec)
ftp>
```

Top

Figure: ftp commands

```
ftp 192.168.36.6
Trying to connect...192.168.36.6
Connected to 192.168.36.6
220- Welcome to PT Ftp server
Username:EmergencyHead
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>get EmergencyHead.txt

Reading file EmergencyHead.txt from 192.168.36.6:
File transfer in progress...

[Transfer complete - 14 bytes]

14 bytes copied in 0.022 secs (636 bytes/sec)
ftp>
```

```
14 bytes copied in 0.022 secs (636 bytes/sec)
ftp>rename EmergencyHead.txt Emergency1.txt

Renaming EmergencyHead.txt
ftp>
[OK Renamed file successfully from EmergencyHead.txt to Emergency1.txt]
ftp>
```

Figure: ftp commands renaming

```
ftp>
[OK Renamed file successfully from EmergencyHead.txt to Emergency1.txt]
ftp>dir

Listing /ftp directory from 192.168.36.6:
0   : Emergency1.txt                                    14
1   : abc.txt                                           73
2   : asa842-k8.bin                                     5571584
3   : asa923-k8.bin                                     30468096
4   : c1841-advipservicesk9-mz.124-15.T1.bin            33591768
5   : c1841-ipbase-mz.123-14.T7.bin                     13832032
6   : c1841-ipbasek9-mz.124-12.bin                      16599160
7   : c1900-universalk9-mz.SPA.155-3.M4a.bin            33591768
8   : c2600-advipservicesk9-mz.124-15.T1.bin            33591768
9   : c2600-i-mz.122-28.bin                             5571584
10  : c2600-ipbasek9-mz.124-8.bin                       13169700
11  : c2800nm-advipservicesk9-mz.124-15.T1.bin          50938004
12  : c2800nm-advipservicesk9-mz.151-4.M4.bin           33591768
13  : c2800nm-ipbase-mz.123-14.T7.bin                   5571584
14  : c2800nm-ipbasek9-mz.124-8.bin                     15522644
15  : c2900-universalk9-mz.SPA.155-3.M4a.bin            33591768
16  : c2950-i6q412-mz.121-22.EA4.bin                    3058048
17  : c2950-i6q412-mz.121-22.EA8.bin                    3117390
18  : c2960-lanbase-mz.122-25.FX.bin                    4414921
19  : c2960-lanbase-mz.122-25.SEE1.bin                  4670455
20  : c2960-lanbasek9-mz.150-2.SE4.bin                  4670455
21  : c3560-advipservicesk9-mz.122-37.SE1.bin           8662192
22  : c3560-advipservicesk9-mz.122-46.SE.bin            10713279
23  : c800-universalk9-mz.SPA.152-4.M4.bin              33591768
24  : c800-universalk9-mz.SPA.154-3.M6a.bin             83029236
25  : cat3k_caa-universalk9.16.03.02.SPA.bin            505532849
26  : cgr1000-universalk9-mz.SPA.154-2.CG               159487552
27  : cgr1000-universalk9-mz.SPA.156-3.CG               184530138
28  : ir800-universalk9-bundle.SPA.156-3.M.bin          160968869
29  : ir800-universalk9-mz.SPA.155-3.M                  61750062
30  : ir800-universalk9-mz.SPA.156-3.M                  63753767
31  : ir800_yocto-1.7.2.tar                             2877440
32  : ir800_yocto-1.7.2_python-2.7.3.tar                6912000
33  : pt1000-i-mz.122-28.bin                            5571584
34  : pt3000-i6q412-mz.121-22.EA4.bin                   3117390
ftp>
```

```
34  : pt3000-16q412-m2.121-22.EA4.bin                    3117390
ftp>delete abc.txt

Deleting file abc.txt from 192.168.36.6: ftp>
[Deleted file abc.txt successfully ]
ftp>
```

Top

3.3 Email Communication:

Each PC was configured with an email client and tested by sending messages across various departments. Messages were successfully delivered and received without delay or error, proving that the email server configuration was effective and reliable.

- **Screenshot:**

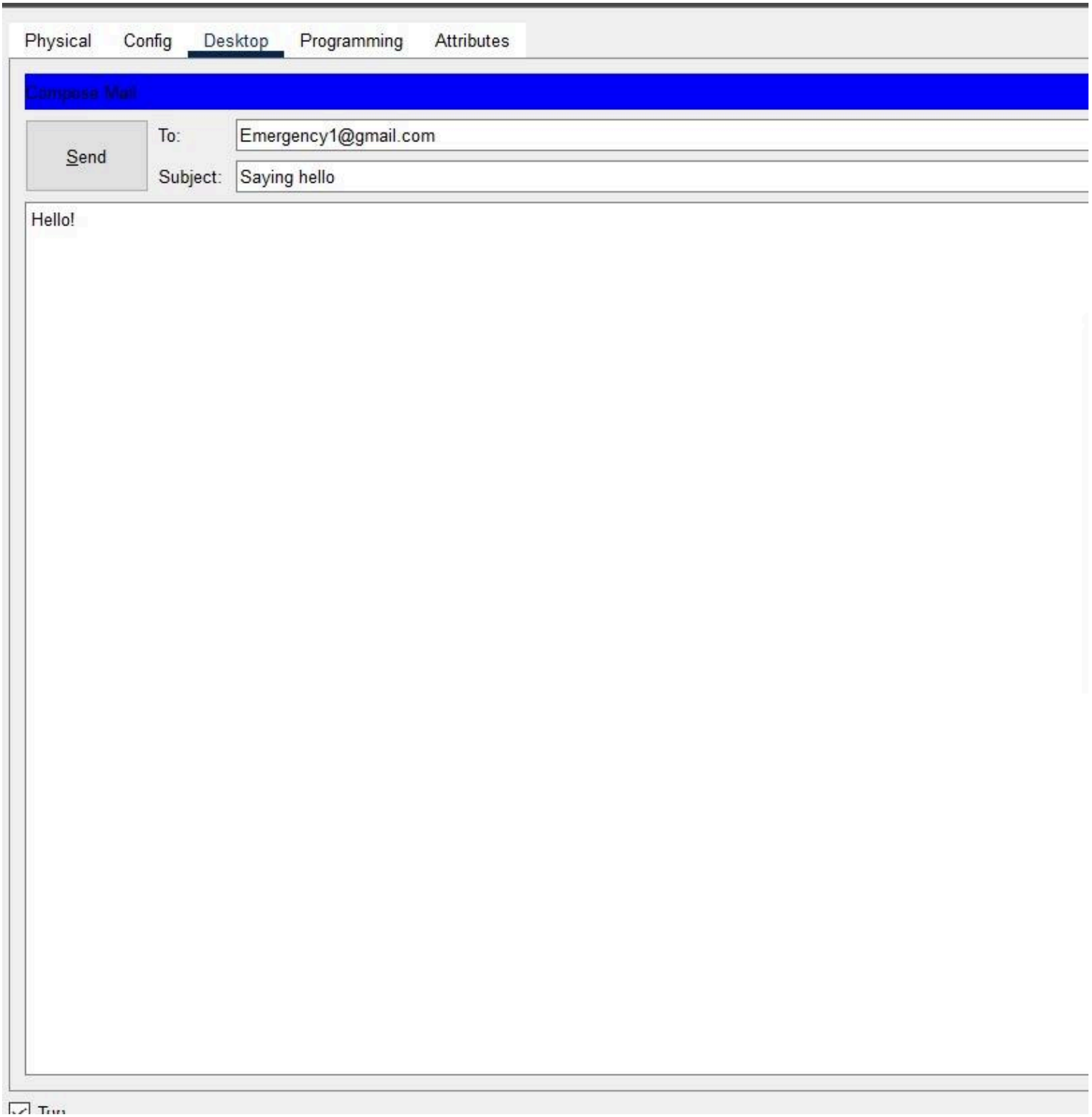

Figure: Sending mail

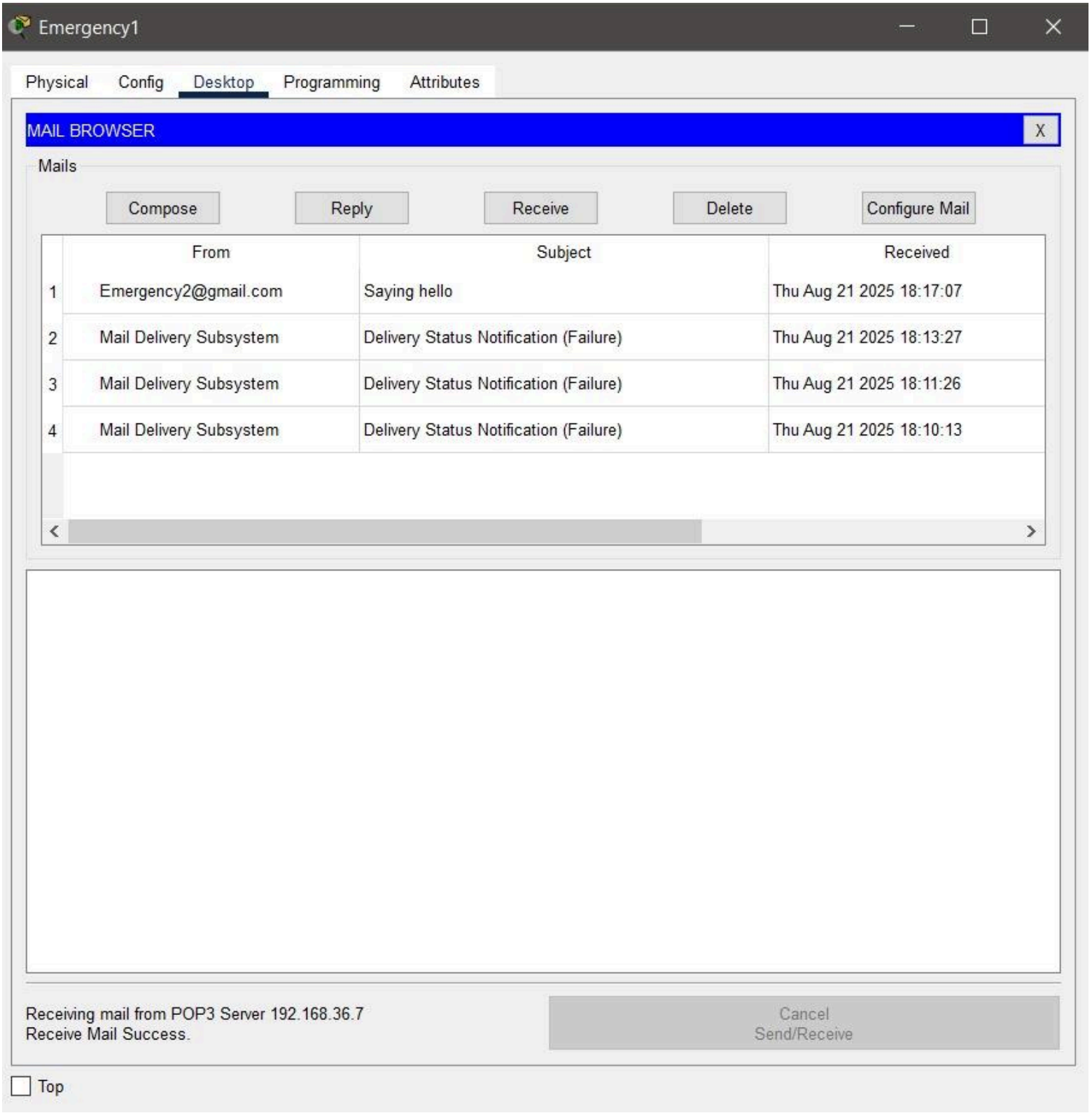

Figure : receiving mail

Chapter 4: Conclusion

To design and implement a secure and scalable hospital-wide network infrastructure for a smart
healthcare system using Cisco Packet Tracer. The network will support patient record management, medical device monitoring, and departmental communication with prioritized Services.
Due to the large number of devices, only sample configuration screenshots are included. However,
all devices were configured with appropriate IP settings, roles, and services as reflected in the simulation file. The complete topology and functional results validate the successful implementation.
The implementation of a network infrastructure for hospital wide network using Cisco Packet Tracer demonstrated key networking concepts and best practices. Each academic and administrative department was designed as a separate LAN with its own subnet, enabling organized communication. These LANs were interconnected through routers using static routing,
allowing seamless data exchange between departments.
The project emphasized logical IP addressing and subnetting, ensuring traffic separation and easier
management.Testing confirmed successful connectivity, service availability, and routing. Overall,
the design offers a scalable, secure, and educationally relevant model for campus network deployment.
Overall, the project achieved its objectives and provided valuable hands-on experience in designing, configuring, and managing a complete enterprise-level network infrastructure